Software-Defined Industrial Visual Anomaly Detection System

Cloud MLOps Pipeline & Simulation

A graduation project that inspects industrial surface images (ceramics and metals), localises defects with bounding boxes and confidence scores, and returns a PASS/FAIL verdict in under one second (warm path). The production line is simulated, so no physical hardware or factory access is needed.

Flow: Python Simulator -> Amazon S3 -> AWS Lambda -> SageMaker Serverless (YOLOv8 ONNX) -> DynamoDB -> Streamlit Dashboard

Project Info

Item	Details
Supervisor	Randa Hesiam
Team Lead	Seif Mohamed
Project start	18 October 2026
Final delivery	28 November 2026 (6 weeks)

Team

Team	Members
Data	Malek Mohamed, Abdelrahman Ramadan
Modeling	Saeed Ali, Mostafa Mohamed
Deployment	Seif Mohamed (Team Lead), Ahmed Khaled

Documentation (Submission 1)

#	Document	Contents
0	Project Proposal	Overview, objectives, scope, datasets, architecture, MLOps loop, milestones
1	01_project_planning_and_management.pdf	Project proposal summary, project plan (Gantt chart, milestones, deliverables, resource allocation), task assignment & roles, risk assessment, KPIs
2	02_literature_review.pdf	Feedback & evaluation, suggested improvements, final grading criteria
3	03_requirements_gathering.pdf	Stakeholder analysis, user stories & use cases, functional and non-functional requirements

Key Technologies

- **Model:** YOLOv8 + ONNX Runtime (production); PatchCore (offline baseline only)
- **Datasets:** MVTec AD, NEU, DAGM, one public Kaggle/Roboflow ceramics set
- **AWS:** S3, Lambda, SageMaker Serverless Inference, SageMaker Model Registry, EventBridge, DynamoDB, CloudWatch, SNS
- **Tools:** Python, PyTorch, Ultralytics, OpenCV, Roboflow, Streamlit, boto3, Git

Milestones

Milestone	Window
M1 Requirements, Data Curation & Environment	18 - 27 Oct
M2 Modelling, Ablations & ONNX Benchmarking	26 Oct - 8 Nov
M3 Serverless Deployment & Latency	5 - 14 Nov
M4 Simulator, Pipeline & Monitoring	8 - 20 Nov
M5 Dashboard, MLOps Loop & Defense	15 - 28 Nov

Target KPIs

- Image-level recall (critical classes) >= 95%
- mAP@0.5 (held-out test set) >= 0.90
- False reject rate <= 5%
- Warm end-to-end latency < 1.0 s
- Audit coverage 100%
- AWS budget: 60 USD

Notes

Datasets are public (MVTec AD is non-commercial research use), so results demonstrate the method and pipeline, not guaranteed performance on a specific production line.